# Clustering II

- Sequential clustering: BSAS and incremental mean.
- Hierarchical clustering: linkage and dendrogram.
- Ward: the connection to k-means.

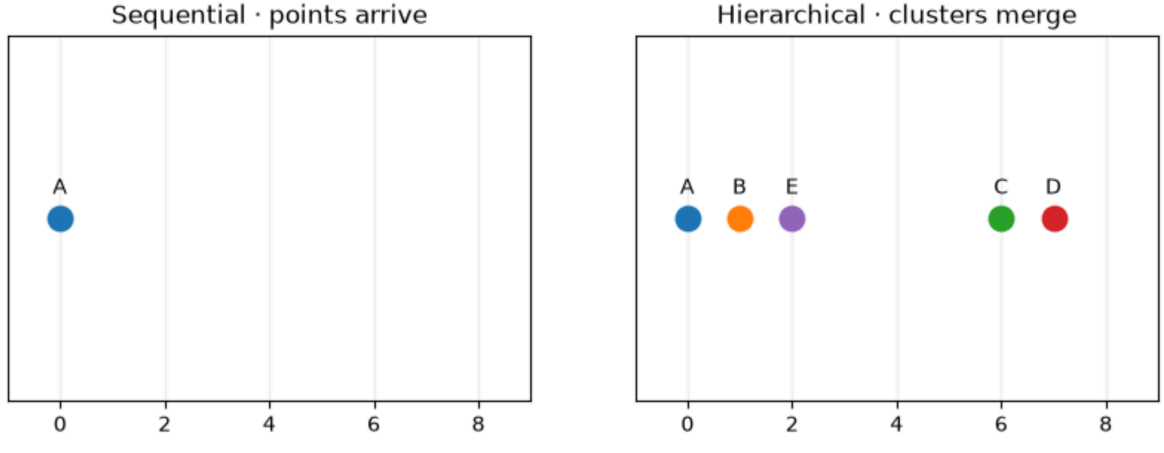
# From batch to sequential

- K-means repeatedly reconsiders all data.
- Sequential algorithms read one point at a time.
- Each new point is assigned immediately or starts a new group.
- Earlier decisions can influence later ones.

## Animated comparison: sequential vs agglomerative

- Left: a new point arrives and joins a group or creates one.
- Right: all points exist from the start; groups are merged.

Advance both algorithms 0



# BSAS: parameters and state

- Distance threshold $\theta > 0$.
- Maximum number of clusters $q_{\max}$.
- Current clusters: $q \leq q_{\max}$ and centroids $\boldsymbol{\mu}_1, \ldots, \boldsymbol{\mu}_q$.

$$\theta > 0, \qquad q \leq q_{\max}$$

# BSAS: assignment rule

- First point forms the first cluster.
- Find nearest existing center and its distance $\delta_i$.
- If too far and room remains, create a new group.
- Otherwise, assign to nearest group and update its mean.

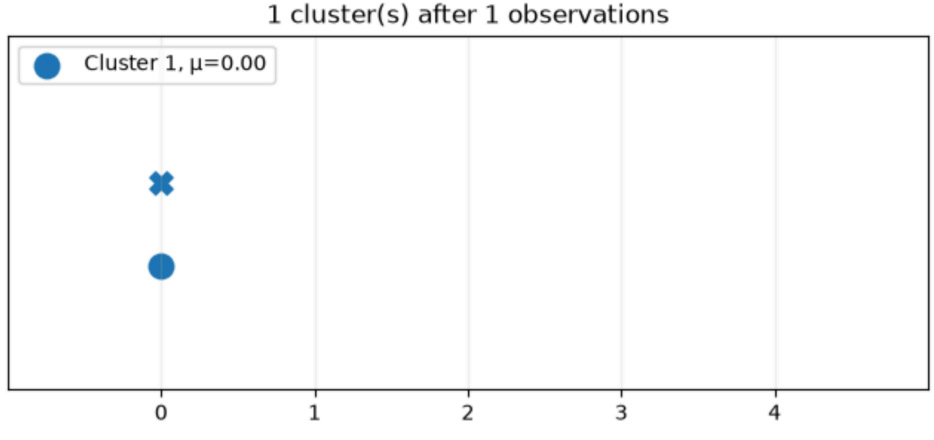
$$r^* = \arg\min_{1 \le r \le q} \|\mathbf{x}_i - \boldsymbol{\mu}_r\|, \quad \delta_i = \|\mathbf{x}_i - \boldsymbol{\mu}_{r^*}\|$$

## BSAS: one point at a time

- Data arrive in the order $0, 2, 3, 4$.
- Change the threshold and advance one observation at a time.

Points received 1

Threshold θ 2.5



# Incremental centroid: no need to reread data

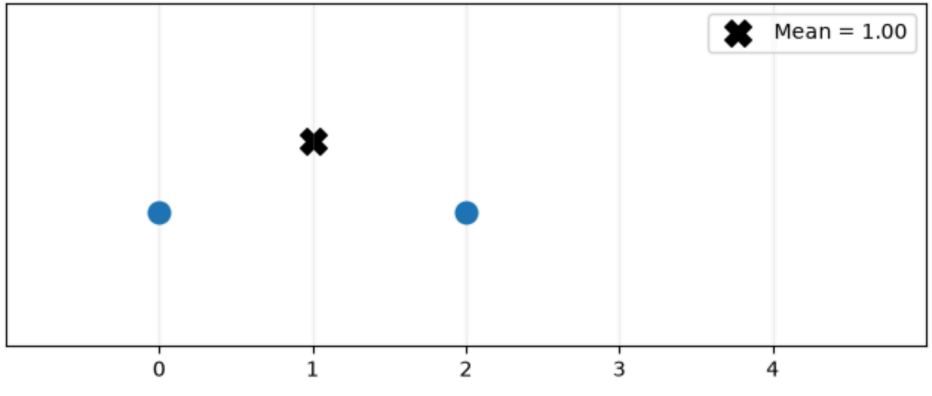
- Suppose a group contains $n_r$ points and has mean $\boldsymbol{\mu}_r$.
- Add one observation $\mathbf{x}_i$.
- The new mean uses only old mean, count and new point.

$$\boxed{\boldsymbol{\mu}_r^{\text{new}} = \frac{n_r \boldsymbol{\mu}_r + \mathbf{x}_i}{n_r + 1} = \boldsymbol{\mu}_r + \frac{\mathbf{x}_i - \boldsymbol{\mu}_r}{n_r + 1}}$$

## Incremental mean: the centroid moves

- Add points to the same group.
- The influence of each additional point becomes smaller as $n$ grows.

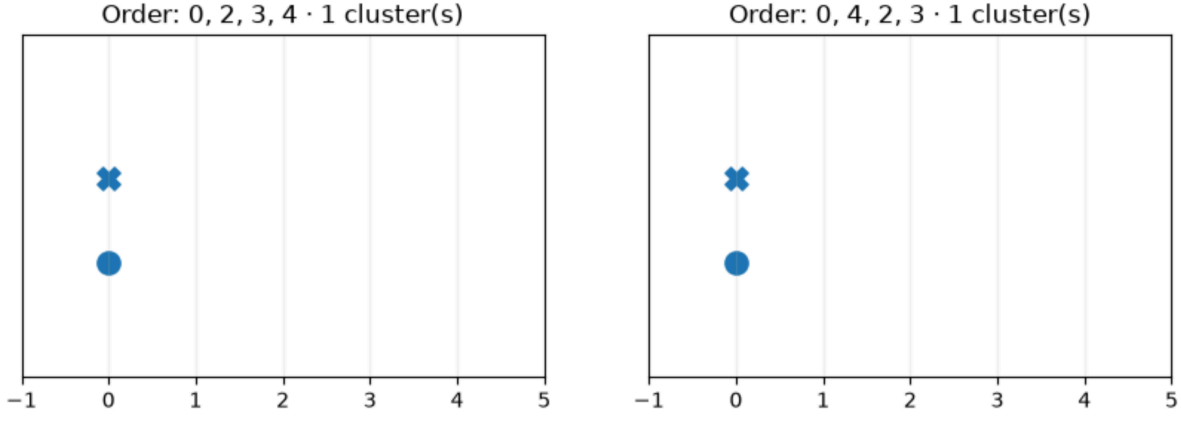
Observations received 2



## BSAS: same observations, different order

- Compare $0, 2, 3, 4$ against $0, 4, 2, 3$.
- With $\theta = 2.5$, one order produces one cluster, the other two.

Observations received 1



# Sequential variants: the idea

- MBSAS: separate seed-selection from final assignment.
- TTSAS: two thresholds; defer ambiguous decisions.
- Order dependence is reduced in some variants, not universally eliminated.

$$0 < \theta_1 < \theta_2$$

# Hierarchical clustering: why a hierarchy?

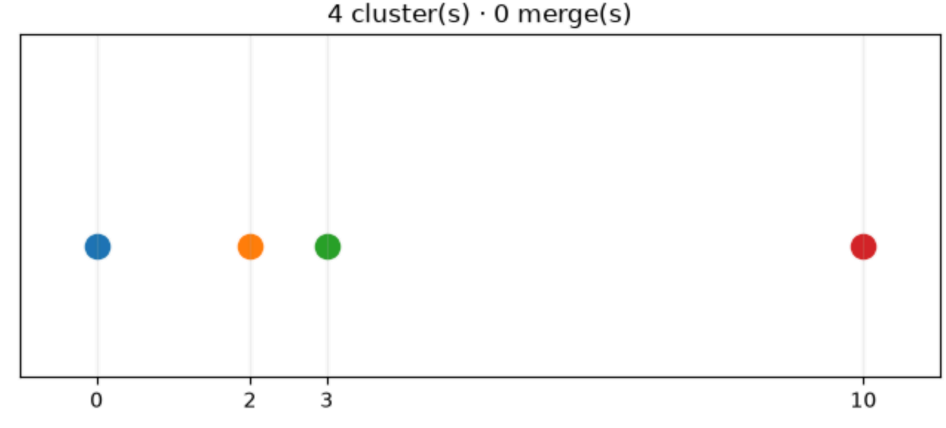
- Rather than commit immediately to a single partition, build nested partitions.
- Agglomerative: begin with $N$ singleton clusters.
- At each step, merge the two closest clusters.
- Merges are irreversible.

$$N \longrightarrow N-1 \longrightarrow \cdots \longrightarrow 2 \longrightarrow 1$$

## Hierarchical merging: watch every fusion

- All four observations exist from the beginning.
- Each step fuses two clusters; none is created by arriving data.

Number of merges 0



# What is linkage?

- Distance between points is not distance between sets.
- Single: closest cross-cluster pair.
- Complete: farthest cross-cluster pair.
- Average: mean over all cross-cluster pairs.

$$D_{\text{single}}(A, B) = \min_{i \in A, j \in B} d(\mathbf{x}_i, \mathbf{x}_j)$$

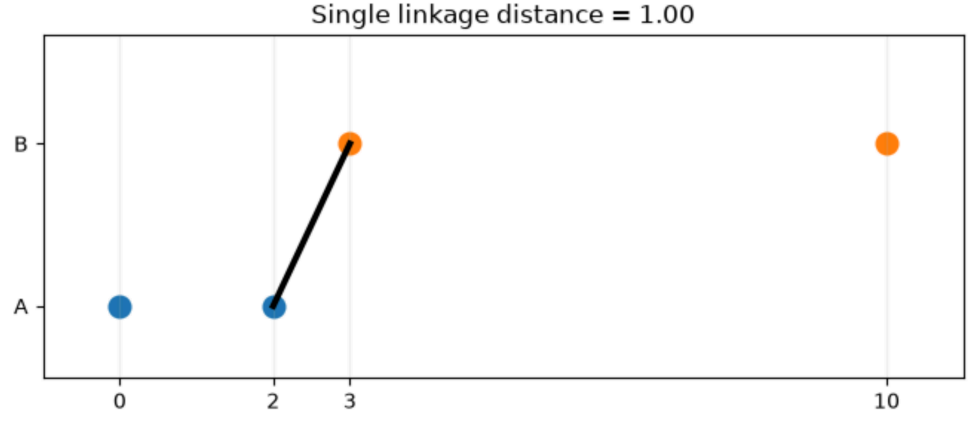
$$D_{\text{complete}}(A, B) = \max_{i \in A, j \in B} d(\mathbf{x}_i, \mathbf{x}_j)$$

## Linkage animation: nearest / farthest / average

- Two existing groups, four cross-group distances.
- The definition of inter-cluster proximity changes with linkage.

$$D_{\text{average}}(A, B) = \frac{1}{|A||B|} \sum_{i \in A} \sum_{j \in B} d(\mathbf{x}_i, \mathbf{x}_j)$$

0: single · 1: complete · 2: average 0



# Agglomerative algorithm

- Initialize $\mathcal{P}^{(0)}$ with $N$ singleton sets.
- Compute inter-cluster linkage distances.
- Merge the pair with smallest linkage distance.
- Repeat until the desired number of clusters (or one).

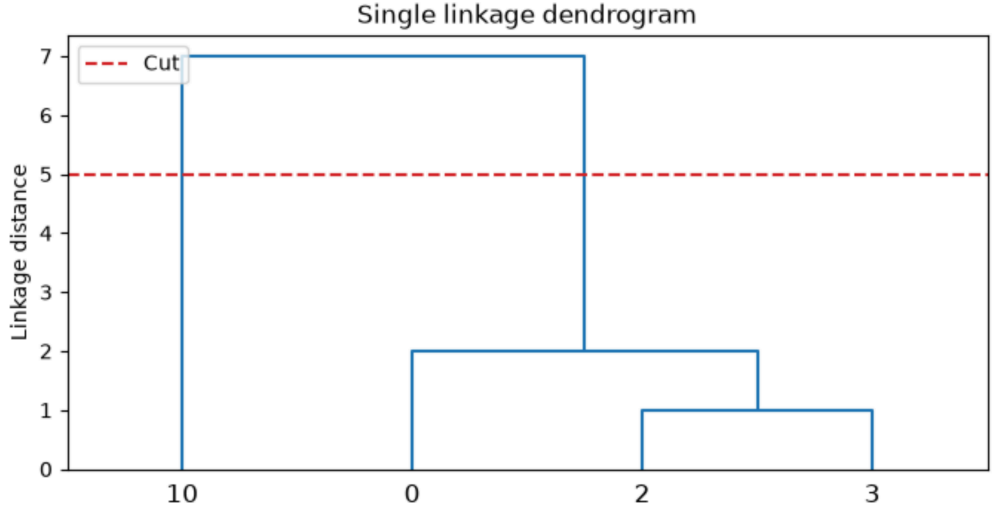
$$(A^*, B^*) = \arg\min_{A \neq B} D_{\text{linkage}}(A, B)$$

## Dendrogram: four points, three linkage choices

- The same merges can occur at different heights.
- Cut the tree to obtain a chosen partition.

0: single · 1: complete · 2: average 0

Cut height 5



# Worked example: dendrogram heights

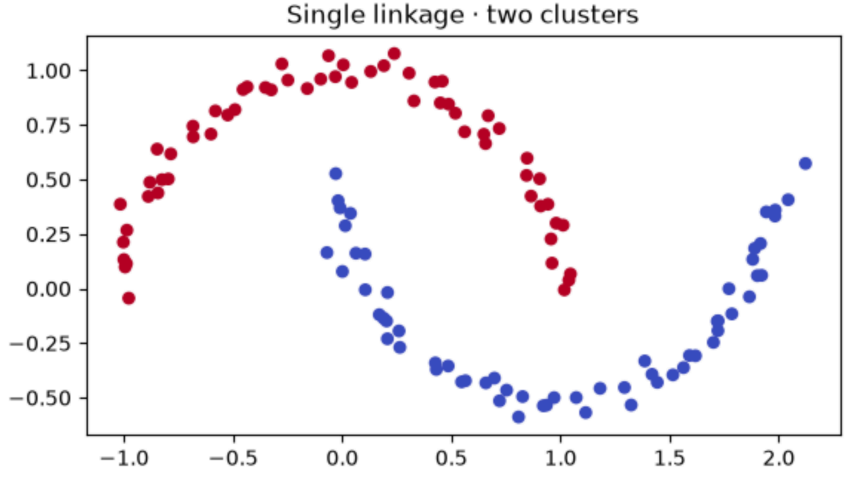
- Observations on a line: $0, 2, 3, 10$.
- First merge: $2$ and $3$ at distance $1$.
- Second merge: $0$ with $\{2, 3\}$.
- Final merge with $10$: heights depend on linkage.

| | single | complete | average |
|---|---|---|---|
| second merge | $2$ | $3$ | $2.5$ |
| last merge | $7$ | $10$ | $25/3$ |

## Two moons: single vs complete linkage

- Single linkage can connect a curved chain.
- Complete linkage tends to favor tighter, smaller-diameter groups.

0: single · 1: complete 0



# Ward: connect hierarchy to k-means

- Instead of closest / farthest pair, measure the increase in squared within-cluster error.
- Merge the pair producing the smallest increase.
- Same squared-error idea, but merges are irreversible.

$$S(A) = \sum_{i \in A} \|\mathbf{x}_i - \boldsymbol{\mu}_A\|^2, \qquad \Delta(A, B) = S(A \cup B) - S(A) - S(B)$$

# Ward: final formula (derive on the board)

- Write the mean of the union.
- Expand squared distances around the two old means.
- Cross-terms vanish because deviations sum to zero.

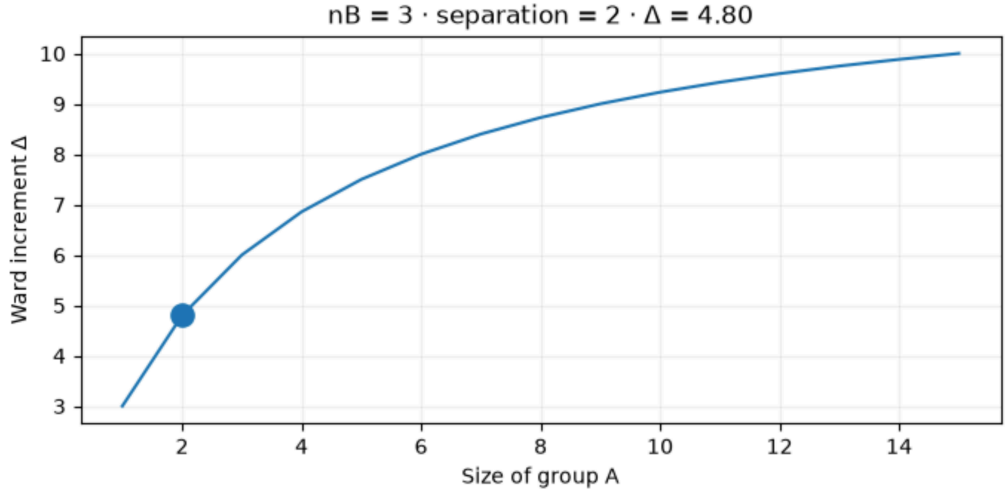
$$\boxed{\Delta(A,B) = \frac{n_A n_B}{n_A + n_B} \|\boldsymbol{\mu}_A - \boldsymbol{\mu}_B\|^2}$$

## Ward: distance and cluster sizes both matter

- Fix centroid separation and change the size of cluster A.
- Ward weighs squared separation by group sizes.

Number of points in A 2

Centroid separation 2



# Ward vs k-means

- K-means: reassign observations and update centroids repeatedly.
- Ward: merge entire groups; cannot undo past merges.
- Both are built around within-cluster squared errors.

$$J_{\text{kmeans}} = \sum_r S(C_r), \qquad \text{Ward: choose } \arg\min_{A,B} \Delta(A, B)$$

# Summary: no universal clustering algorithm

- K-means: compactness around centers.
- Sequential BSAS: decisions upon arrival; order matters.
- Hierarchical: nested structure via irreversible merges.
- Single / complete / average / Ward express different notions of group.

# Final takeaway

- Start with the meaning of proximity.
- Then choose what a cluster should represent.
- Only then choose the algorithm.

$$\text{proximity} \longrightarrow \text{cluster notion} \longrightarrow \text{algorithm}$$